

RESONANCE R14O USER HANDBOOK MANUAL

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OAE Screener

QUICK USER'S HANDBOOK



r140



resonance 

Release history

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First issue

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added DPOAE Screener

M.R.S. s.r.l. is the manufacturer of the RI4O OAE Screener.



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1. Introduction and intended use

RESONANCE thanks you for purchasing one of its medical devices from its portfolio of products.

This manual contains information and warnings aimed at ensuring the correct and safe use of the r14o OAE Screener available in the following versions:

Description	Product code
r14o - TE screener	4300103010
r14o - TE advanced screener	4300103030
r14o - DP screener	4300103015
r14o - DP advanced screener	4300103040
r14o-TE/DP advanced screener	4300103050

Carefully store this manual away in a safe place for each future consultation.

The r14o OAE Screener is designed and manufactured to meet or exceed all quality and safety requirements, and has been certified with CE-symbol according to European Medical Devices Directive. All the functions are software-controlled, upgrading the software at a later date will be simple and free of charge.

The r14o OAE Screener is intended to be used by audiologists, hearing healthcare professionals, or trained technicians.

The Otoacoustic Emissions (OAE) screener is a device testing the cochlear functionality which relates to the amplification activity by the outer hair cells. When the cochlea is stimulated by sound the outer hair cells vibrate producing an almost imperceptible sound which goes back to the middle-ear. This subtle sound is measured by a probe inserted into the ear canal.

The user must therefore have an adequate knowledge of audiological diagnostic equipment, expertise on the examination methods used and knowledge of English.

This device is intended to be used in medical environments with low levels of background noise. The operator must exercise caution in interpreting results where background noise has not been sufficiently isolated.

It is recommended that the device be operated within an ambient temperature range of 15 -35 degree Celsius.



This operating manual is not intended as a training manual for audiometry and tympanometry. The reader should consult standard audiology texts for the theory and application of the screening tests provided by this device.

2. Description

2.1 Precautions



WARNING indicates a hazardous situation which, if not avoided, could result in death or serious injury.



NOTICE is used to address practices not related to personal injury.

This instrument has been built and tested in compliance with product standards and left the production factory in perfect technical safety conditions.



- Prior to testing, the performance of the probe should be checked in the dedicated test cavity by the operator.
- Ensure that the correct tip size is selected to give a good seal in the ear canal. Use only the single use ear tips provided with the unit. Replacement tips are available from RESONANCE or one of their agents.
- The device transducers and parts in contact with the patient should be cleaned following every use. Appropriate antiseptic solution should be used in compliance to the maintenance and cleaning section of this manual.
- The device is intended to be powered from Li-ion internal rechargeable battery or from a medical CE power supply. Use of power supplies other than those provided by the manufacturer is not permitted. When used, position so that the power plug can be easily disconnected.
- Although the instrument fulfils the relevant EMC requirements, precautions should be taken to avoid unnecessary exposure to strong electro-magnetic fields, e.g., from mobile phones, microwaves, radiotherapy devices, etc.; these may induce malfunctions in their performance. If the device is used adjacent to other equipment it must be observed that no mutual disturbance appears.
- r14o does not have specific ingress protection. Care should be taken to avoid liquid ingress. Immersion of the device may result in serious damage.

Any possible modification and/or alteration to this device is prohibited, without prior authorization thereto by the Manufacturer.



If the r14o is connected to other devices with their own power source, the system conforms to IEC 60601-1 must be checked under the responsibility of the installer; in the event that isolation devices are used to achieve such conformity, these must comply with IEC 60601-1. Any person who connects external equipment to the device connectors has formed a system and is therefore responsible for the system to comply with the requirements of IEC 60601-1.

If in doubt, contact your RESONANCE technician or local distributor for help.



Device not to be used in areas in which flammable anesthetics or flammable agents for disinfection or skin cleaning are applied.



Electric and electronic waste may contain hazardous substances and therefore must be disposed of separately.

Such products will be marked with the crossed-out wheeled bin shown.

The cooperation of the user is important in order to ensure a high level of reuse and recycling of electric and electronic waste.

Failure to recycle such waste products in an appropriate way may endanger the environment and consequently the health of human beings.

2.2 Warranty and maintenance

RESONANCE warrants that:

- The r14o is free from defects in material and workmanship under normal use and service for a period of 24 months from the date of device's installation by authorized personnel to the first purchaser.
- Accessories are free from defects in material and workmanship under normal use and service for a period of 12 months from the date of device's installation by authorized personnel to the first purchaser.

Furthermore, this warranty shall not apply to, and M.R.S. shall not be responsible for, any loss arising in connection with the purchase or use of any RESONANCE device that has been:

- repaired by anyone other than an authorized RESONANCE service representative;
- altered in any way so as, in RESONANCE judgment, to affect its stability or reliability;
- subject to misuse or negligence or accident, or which has had the serial number altered, effaced or removed; or improperly maintained; or used in any manner other than in accordance with the instructions furnished by RESONANCE.
- This warranty is in lieu of all other warranties, express or implied, and of all other obligations or liabilities of M.R.S., and M.R.S. does not give or grant, directly or indirectly, the authority to any representative or other person to assume on behalf of M.R.S. any other liability in connection with the sale of RESONANCE products.
- The r14o may be repaired only by your dealer or by a service company authorized by RESONANCE, recommended by your dealer.
- Only original RESONANCE spare or consumable parts must be used for service or repair the r14o.

In order to ensure that your device works correctly, the r14o should be checked and calibrated at least once a year. We recommend to send the r14o to your RESONANCE dealer for this service, including the probe and all of the accessories. The verification of instrument calibration must be conducted by qualified service personnel authorized by the manufacturer via appropriate instrumentation. This must provide the level control and the frequency of all signals emitted by the transducers supplied with the instrument.

Should transducers require replacement, use only parts provided by the manufacturer. Replacement of transducers will require re-calibration of the instrument.

2.3 Unpacking and checking the r14o

On checking for packaging and content damage immediately after the device’s reception; should any sign of damage or tampering be noted, please notify the carrier immediately.

If the device has been damaged during transportation, the device should be checked for any electrical or mechanical defects.

Should the device be dropped or fall, it may sustain hidden damage that may result in hazardous operation.

In case of any doubt, please do not use the device and either contact your Resonance Dealer, or e-mail the Resonance Customer Support Service directly at support@resonance-audiology.com



Keep all original packaging for future use.
The identification label on the rear plate holds the serial number. This should be checked and written down for later service claims.

2.4 Accessories

List of accessories	Otoacoustic emission probe	Calibration cavity	
Kit Tips for OAE probe			
Medical CE approved power supply			Cradle
Multilingual CE Operation manual			MDS Software
USB pen drive including: DHR documents, MDS software installer, Multilingual CE Operation manual			
Carrying bag			
Optional available	External USB Keyboard	Wi-Fi thermal printer	
Available tests			
Otoacoustic emission			
*TEOAE screening			
*TEOAE diagnostic			
*DPOAE screening			
*DPOAE diagnostic			

* Optional test, depending on device model

Accessories

Image	Description	Standard / Optional	Plug
	OAE Probe*	•	PROBE (3)
	OAE Test cavity	•	/
	Kit Tips OAE	•	/
	Medical CE approved power supply: Model: MENB1010A0503(xyz) (xyz) = B01 changeable direct plug-in (xyz) = Q01 with power cord (xyz) = N01 with power cord	•	5VDC (1)
	As alternative: Medical CE approved power supply: Model: ME10A0503(xyz) (xyz) = B01 changeable direct plug-in (xyz) = Q01 with power cord (xyz) = N01 with power cord	•	5VDC (1)
	Charging Cradle	•	/

	USB pen drive	●	for PC use
	Carrying bag	●	/
	Wi-Fi thermal printer	○	/
	External USB Keyboard	○	 (2)

● = standard

○ = optional

* = applied part

The applied parts are represented by the transducers and pushbutton which are placed in contact with the patient. They are indicated by an asterisk (*).

2.5 Connecting the accessories



Please connect all the cables before powering on the device.



Reference	Symbol	Function
1	5VDC	Socket for external power supply
2		USB Host port for connection of pen drive (for software update) or external keyboard (optional)
3	Probe	Socket for Probe: OAE probe
4		USB Slave connector: used for wired data transfer
5 - 6	n.a.	Not used

2.6 Power Sources

The device can operate with its own internal rechargeable battery (Li-ion 3.6 V / 3 Ah), or connected to the medical power supply, or placed on the charging cradle.

The average recharge time of an empty battery is less than 4 hours, both with direct feeder link or with its charging cradle. Two hours are enough to reach 80% charge.

To increase the battery duration, the display brightness is reduced to 20% after 60 seconds of inactivity (no action on the touchscreen).

After 2 minutes of inactivity starting from the reduction of display brightness, the device switches to sleep mode. Simply touch the screen to reactivate the device. After 8 minutes of inactivity, and if not connected to the battery charger, the instrument switches off automatically. Press the ON / OFF button to restart the device.

2.7 Installation

Ensure all transducers are connected prior to starting the device.

Turn the device on and check the battery status: at first switching on, the battery indicator may provide an approximate value. A full charge cycle is suggested prior to its use.

2.8 Battery indicator

The battery charge status is showed on the display with a dedicated icon; a numeric indicator shows the percentage of remaining battery autonomy.

When less than 10%, the battery indicator flashes to underline that recharging is necessary.

At low charge levels, the device displays a **Low Battery** warning message. Some tests may not be available following the low-battery warning.



Although the device is able to operate even when the battery is running very low, it is not recommended to use it with less than 5% battery autonomy and it is recommended to connect it to the medical power supply or to its charging cradle.

2.9 Symbols present on device

	5VDC - Direct current (Medical CE approved power supply)
	Refer to instruction
	Refer to manual for transducer and power supply connection
	Class II device. (Symbol refers to external medical power supply with double insulation)
	BF type applied parts
	Electric and electronic waste may contain hazardous substances and therefore has to be collected separately
	Together with name and address indicates the manufacturer of the instrument
	Conformity marks to directive 93/42/CEE - notification body 0051 (IMQ)
	Year of production of the device
	USB Host port
	USB slave port

3. Components and connectors



	Description
1	Socket for Probe
2 - 3	Not used
4	7" TFT Resistive touchscreen display
5	Main switch
6	Led: shows that battery is charging
7	Socket for medical power supply
8	USB Slave connector: used for wired data transfer
9	USB Host port for connection of pen drive (for software update) or external keyboard (optional)

3.1 Power on and off the device

To turn on the device, press briefly the Main switch (5) and wait for the appearance of the bootscreen on display.

To switch off the device, press the Main switch (5) for about 2" and then release the button.



Before powering on the device, ensure that all the accessories are already plugged-in.

3.2 PIN Code request

When switching the device on a PIN code is requested to access both the device functionalities and the data it contains. The default PIN code set by the manufacturer is 1234. After the PIN code has been typed in the starting screen is presented. The PIN code can be changed or disabled through the dedicated function available in the setup menu. For additional details see chapter 5.1



In case wrong codes are typed in, after 3 failures the possibility of new attempts is prevented, and unlocking is only possible with the PUK code provided along with the device. Should the PUK code be lost, it can be retrieved by communicating to the manufacturer the device serial and product numbers to be found on the label on the lower side of the unit.

4. Performing tests

4.1 TEOAE Screening

The Otoacoustic Emissions Test allows evaluating outer hair cell functionality. The test is performed using a small probe which is inserted into the external ear canal, stimulating the cochlea by a wideband signal and measuring the response signal produced by the outer cells. The automatic screening result is based on the signal to noise ratio of the otoacoustic emissions.

Test performance



Switch on the device on and press **TEOAE-S** (see picture beside) on the main page to open the test function (see picture below).



The main screen displays bar diagrams for both ears. The ear that is currently selected is highlighted above the diagrams.



To switch the ear to be tested, press the corresponding bar diagram.



Press **START** to start the test.



The bar below the graph shows the test progress.

When the test is finalized, the results are shown on the screen:



PASS: test passed, valid OAE response detected.

REFER: test failed, no OAE response detected.

NOT COMPLETED: test stopped by the operator.



Press **SETUP** (see picture beside) to open the test setup page. The screening setup page is protected with a PIN code. The default PIN for entering the setup is 9999.

The menu allows for the setting of a few test parameters:

The screenshot shows a dark blue setup menu. It has several sections: 'Max Stimuli' with a value of '1250 (75 sec)', 'Artefact Level' with a value of 'OFF', 'Stimulus Level' with a value of '85 dB peSPL', and 'Number Pass' with a value of '3'. On the right, there are three buttons labeled '1', '2', and '3'. At the bottom, there is a table for 'Frequency (KHz)' and 'SNR dB'. The 'Frequency (KHz)' row has values 1, 1.5, 2, 3, 4, and a 'Correlation' column. The 'SNR dB' row has values 3, -, 6, 6, 6, and OFF. The top left shows a back arrow, and the top right shows a battery icon and the time 17:00.

Adjustable parameters:

Max Stimuli

Max test time. By touching this field the maximum test time, between 30 and 75 seconds, can be selected. When a valid result is obtained the test is automatically stopped.

Artefact Level

In this field the maximum acceptable noise level can be selected. When the noise exceeds the set value, the measurement is rejected (artefact). By touching this field a value between 20 and 60 dB SPL can be selected or can be switched off.

SNR dB

By clicking the SNR labels, the signal to noise ratios are set for the individual frequencies. It allows choosing -, 3, 6, 9, and 12 dB. When choosing '-', the frequency is excluded from the screening evaluation.

Number Pass

The overall screening result will require at least this number of frequencies to pass their individual SNR criterion.

Correlation

Allows for setting the minimum required correlation for the overall screening result to be 'PASS'. (See TEOAE Diagnostic for further explanations.)

A maximum of 3 protocols of predefined values can be saved by keeping the numbered fields on the right side of the screen pressed for a few seconds and until they turn green.

4.2 TEOAE Diagnostic

In addition to the "pass" or "refer" result, the diagnostic TEOAE test provides information on the intensity and the frequency where a response has been detected.

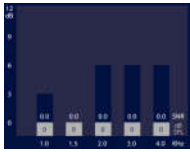
Test performance



Switch the device on and press **TEOAE-D** (see picture beside) on the main page to open the test function.



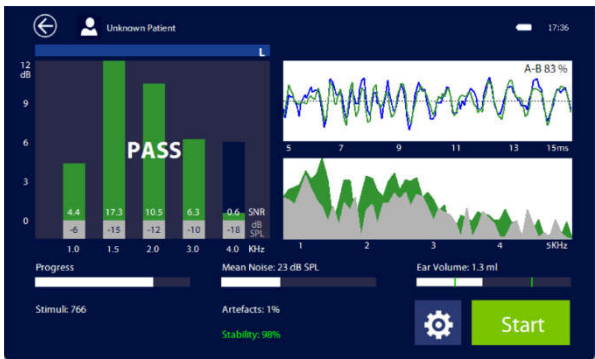
The main screen displays three graphs: on the left side a histogram showing the signal/noise ratio for all frequencies, on the upper right the A and B buffers and on the lower right the signal and noise frequency response.



Press the left graph to switch ear side.



Press **START** to start the test.



All results are displayed on the screen when the test is finalized:

PASS: test passed; the measurement met the set criteria. Detection criteria can be customized in the setup menu, based on the specific screening or diagnostic needs.

REFER: test failed, no OAE response detected. The measurement did not meet the set criteria.

NOT COMPLETED: test stopped by the operator.

On top of the histogram bars the SNR values for the different frequencies are shown.

Stability: this is an indication of the stimulus signal stability. The value indicates the percentage of valid stimuli on the total.

Artefact: this is an indication of the signal quality assessed during the test. This percentage is calculated on the number of acquisitions during which noise has been higher than the setup threshold level.

Noise: test average noise.

The noise, progress and sweep indicators are updated in real time while the test is in progress.



Press **SETUP** (see picture beside) to open the test setup page.

The menu allows for the setting of a few test parameters:



Adjustable parameters:

Max Stimuli

Max test time. By touching this field the maximum test time, between 30 and 75 seconds, can be selected. When a valid result is obtained the test is automatically stopped.

Artefact Level

In this field the maximum acceptable noise level can be selected. When the noise exceeds the setvalue, the measurement is rejected (artefact). By touching this field a value between 20 and 60 dB SPLcan be selected or can be switched off.

SNR dB

By clicking the SNR labels, the signal to noise ratios are set for the individual frequencies. It allows choosing -, 3, 6, 9, and 12 dB. When choosing '-', the frequency is excluded from the screening evaluation.

Number Pass

The overall screening result will require at least this number of frequencies to pass their individual SNR criterion.

Correlation

Allows for setting the minimum required correlation for the overall screening result to be 'PASS'. (See TEOAE Diagnostic for further explanations.)

A maximum of 3 protocols of predefined values can be saved by keeping the numbered fields on the right side of the screen pressed for a few seconds and until they turn green.

4.3 DPOAE Screening

The DPOAE (Distortion Product Otoacoustic Emissions) test is performed by simultaneously presenting two tonal stimuli to which a normal cochlear function responds is the generation of other tonal responses. The size of these responses is plotted in a graph as function of one of the stimulus frequencies and is called a DP-gram.

Otoacoustic emissions prove a good mechanical functionality of the outer hair cells but does not correlate well to inner hair cell functionality.

Normal DPOAEs can be detected in individuals with hearing threshold up to 35 dB HL.

Test performance



Switch the device on and press **DPOAE-S** (see picture beside) on the main page to open the test function.



The main screen displays bar diagrams for both ears. The ear that is currently selected is highlighted above the diagrams.



To switch the ear to be tested, press the corresponding bar diagram.

Start

Press **START** to start the test.

When the test is finalized, the results are shown on the screen:



PASS: test passed, valid OAE response detected.

REFER: test failed, no OAE response detected.

NOT COMPLETED: test stopped by the operator.



Press **SETUP** (see picture beside) to open the test setup page. The screening setup page is protected with a PIN code. The default PIN for entering the setup is 9999.

The menu allows for the setting of a few test parameters:



Adjustable parameters:

Timeout

Maximum search time per single frequency. By touching this field the maximum waiting time, between 2 and 16 seconds, can be selected. When selecting "Auto" the test proceeds to the next frequency on reaching the pass criterion for the current frequency.

Stimulus Level F1

This field allows to select the pure tone signal intensity for F1.

Possible options:

L1 = L2, the level of F1 will be the same as the level set for F2

L1 = L2 + 10 dB, the level of F1 will be the same as the level set for F2 plus 10 dB

Auto: the level of F1 will be determined based on the following formula: $L1 = 0.4 L2 + 39 \text{ dB}$.

When selecting this option, the stimulus level is automatically adjusted.

Stimulus Level F2

This field allows to select the pure tone signal intensity for F2, between 30 and 70 dB SPL.

Artefact Level

In this field the maximum acceptable noise level can be selected. When the noise exceeds the set value, the measurement is rejected (artefact). By touching this field, a value between 20 and 40 dB SPL can be selected.

Frequency retest

When enabled, it tests again the specific frequency, by slightly change the value, in case of "refer".

Number of frequencies per pass

Minimum number of frequency bands that must exceed the thresholds SNR value set to obtain the test pass.

Frequencies and SNR

The value of the signal to noise ratio for each frequency can be set between 3, 6, 9 or 12 dB, or the frequency band can be disabled by setting "-".

A maximum of 3 sets of predefined values can be saved by keeping the numbered fields on the right side of the screen pressed for a few seconds and until they turn green.

In addition to the 3 user-definable protocols, there are 2 presets: 55/65 and 70/70. These are used for screening tests, perform 5-frequency checks at fixed levels, and cannot be edited.

4.4 DPOAE Diagnostic

The DPOAE (Distortion Product Otoacoustic Emissions) test is performed by simultaneously presenting two tonal stimuli to which a normal cochlear function responds is the generation of other tonal responses. The size of these responses is plotted in a graph as function of one of the stimulus frequencies and is called a DP-gram.

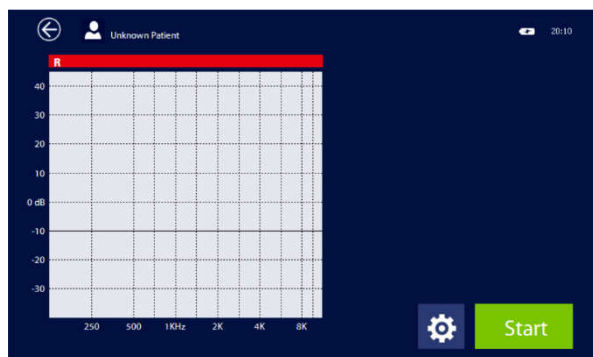
Otoacoustic emissions prove a good mechanical functionality of the outer hair cells but does not correlate well to inner hair cell functionality.

Normal DPOAEs can be detected in individuals with hearing threshold up to 35 dB HL.

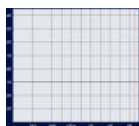
Test performance



Switch the device on and press **DPOAE** (see picture beside) on the main page to open the test function.



The main screen displays a DP-gram where the x-axis represents the reference frequency (f_2) and the y-axis the detected OAE level expressed in dB SPL.

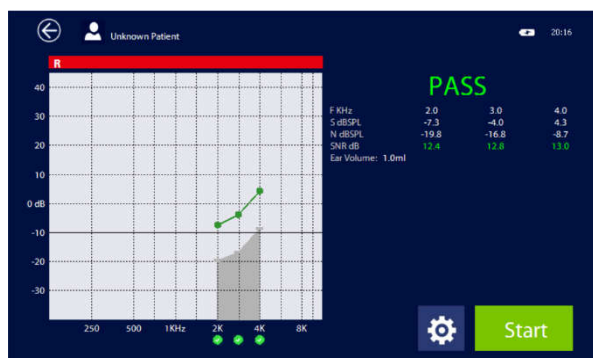


To switch the ear to be tested, press the DP-gram.

Start

Press **START** to start the test.

The test results are displayed on the screen when the test is finalized:



PASS: test passed; the measurement met the set criteria. Detection criteria can be customized in the setup menu, based on the specific screening or diagnostic needs.

REFER: test failed, no OAE response detected. The measurement did not meet the set criteria.

NOT COMPLETED: test stopped by the operator.

Otoacoustic emissions are represented by the green dots and noise by the border of the gray area. When the signal to noise ratio is higher than the predefined threshold the result is “good” and a green marking with checkmark is shown below the corresponding frequency; otherwise, the result is “not good” and a red marker with a cross is shown. To obtain a “good” result, the signal level must exceed the threshold value of -10 dB SPL.

The table on the right shows the numerical presentation of the results in the DP-gram.



Press **SETUP** (see picture beside) to open the test setup page.

The menu allows for the setting of a few test parameters:

Timeout: Auto

Stimulus Level F1: F2 + 10dB SPL

Frequency retest: No

Artefact Level: 40 dB SPL (loud)

Stimulus Level F2: 55 dB SPL

Number Pass: 3

FREQUENCY (kHz)											
1.5	2.0	2.5	3.0	3.3	4.0	4.5	5.0	5.6	6.0	6.5	7.0
3	6	-	6	-	6	-	-	-	-	-	-
SNR dB											

Buttons: [1] [2] [3] [65 / 55] [70 / 70]

Adjustable parameters:

Timeout

Maximum search time per single frequency. By touching this field the maximum waiting time, between 2 and 16 seconds, can be selected. When selecting "Auto" the test proceeds to the next frequency on reaching the pass criterion for the current frequency.

Stimulus Level F1

This field allows to select the pure tone signal intensity for F1.

Possible options:

L1 = L2, the level of F1 will be the same as the level set for F2

L1 = L2 + 10 dB, the level of F1 will be the same as the level set for F2 plus 10 dB

Auto: the level of F1 will be determined based on the following formula: $L1 = 0.4 L2 + 39$ dB. When selecting this option the stimulus level is automatically adjusted.

Stimulus Level F2

This field allows to select the pure tone signal intensity for F2, between 30 and 70 dB SPL.

Artefact Level

In this field the maximum acceptable noise level can be selected. When the noise exceeds the set value, the measurement is rejected (artefact). By touching this field a value between 20 and 40 dB SPL can be selected.

Frequency retest

When enabled, it tests again the specific frequency, by slightly change the value, in case of "refer".

Number of frequencies per pass

Minimum number of frequency bands that must exceed the thresholds SNR value set to obtain the test pass.

Frequencies and SNR

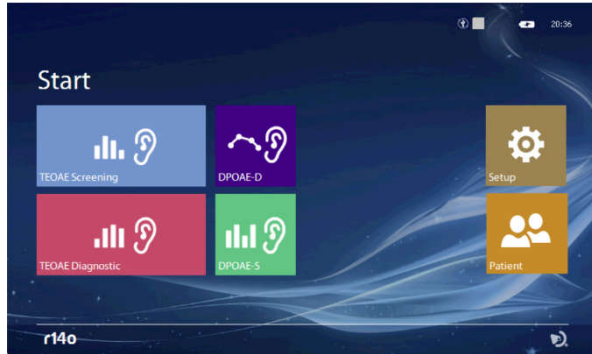
The value of the signal to noise ratio for each frequency can be set between 3, 6, 9 or 12 dB, or the frequency band can be disabled by setting "-".

A maximum of 3 sets of predefined values can be saved by keeping the numbered fields on the right side of the screen pressed for a few seconds and until they turn green.

In addition to the 3 user-definable protocols, there are 2 presets: 55/65 and 70/70. These are used for screening tests, perform 5-frequency checks at fixed levels, and cannot be edited.

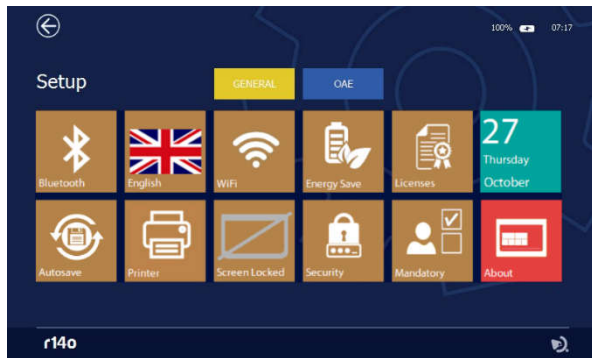
5. Setup settings

After switching on the device, set the parameters that affect the test operation.



Press the **SETUP** icon (see adjacent image) to enter the setup page, there the settings are grouped as follow: GENERAL; OAE.

5.1 General Setup



Press one of the tabs to access the respective Setup page:



Bluetooth



Press the **BLUETOOTH** icon (see adjacent image) to activate database data transmission to PC, if set to OFF transmission is disabled.

Autosave



Press the **AUTOSAVE** icon (see adjacent image) to enable the autosave mode. If enabled, once selected a patient, every performed test will be automatically stored into the data- base after pressing the **EXIT** icon; if disabled a pop-up window will be displayed asking for confirmation.

Language



Through the **LANGUAGE** button (see adjacent image) you can select the interface language. By pressing this icon, a menu with all the available languages will be opened.

Rotation block



Some of the tests provide for their visualization either horizontally or vertically: the visualization is automatically set by rotating the instrument. This option allows to block the rotation keeping the display horizontal.

WiFi



By pressing the **WIFI** icon (see adjacent image) a window allowing the connection to a Wi-Fi network will. This icon is active only when a Wi-Fi USB dongle is inserted in the USB host port. This option can be used to check the connection with the wireless printer.

Power management



In order to improve the battery operating autonomy, you can set 3 different power modes: energy saving, balanced mode and performing mode. Based on the selected mode the display backlight and the unit automatic switching off will change. When the unit is connected to the power supply and the performing mode is selected, the display remains constantly backlit.

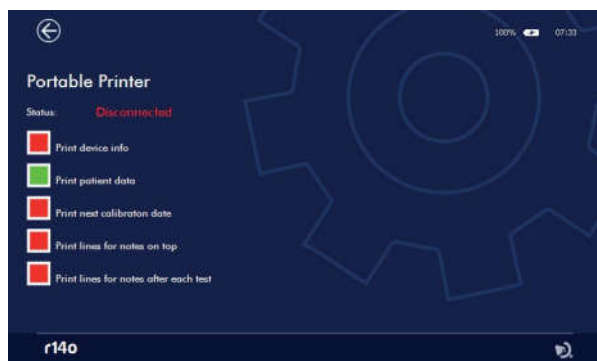
Print-out



By pressing the **PRINT** icon (see adjacent image) you can set the print options for the optional portable printer.

You can select the printing-out of the following fields:

- Unit serial number and logo
- Patient's data
- Calibration expiry date
- Lines for comments after the heading
- Lines for comments after every test



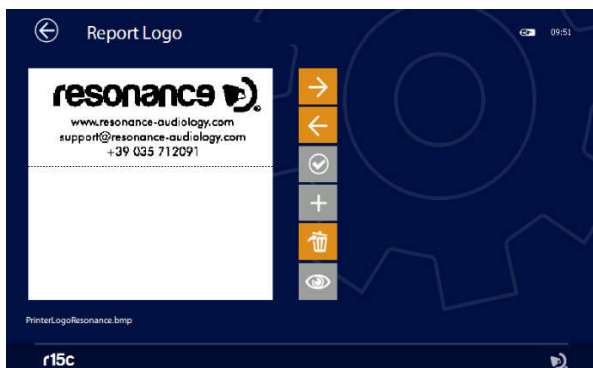
Report header customization

You can customize the report header you print by importing a graphic file that contains your logo or information about your company. The graphic file must be in BMP format, black and white (1 bit x pixel) and have a maximum size of 600x400 pixels. Create a folder named Printer Logo in the root of the pen drive, copy the file or files to be used inside and insert it in the USB HOST port of the device.

It is necessary to copy the file to the device memory to be able to use it for printing.



Pressing the LOGO button (see adjacent image) will open the page for selecting the file



Using the ARROW keys, it is possible to choose the graphic file from those present both in the device and in the pen drive



If the selected file is on the pen drive, the + key will turn orange, pressing it will copy it to the device memory.



Once the desired file has been selected, press the CONFIRM key (see adjacent image) to set it as the print header.



The DELETE button will allow you to delete the selected file.



If the printer is connected, using this key it is possible to print the test page with its logo.

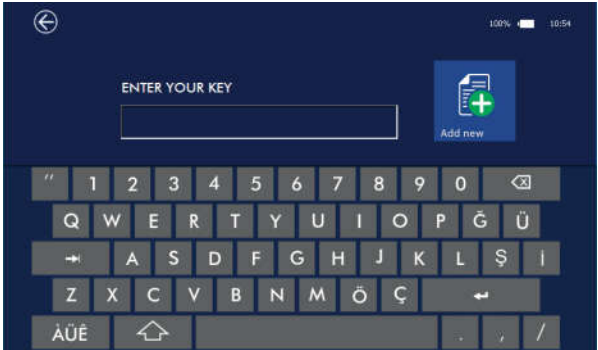
Licenses



Press the LICENSES icon (see adjacent image) to check the active test licenses. The following is displayed:

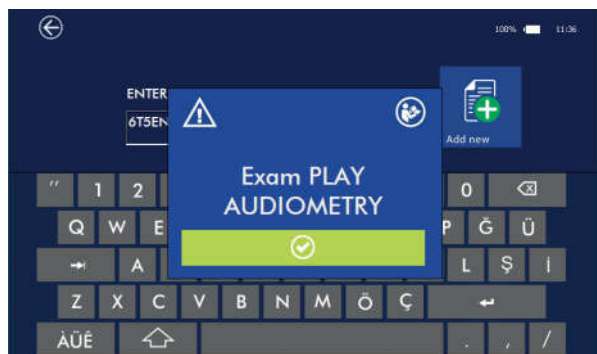


To add a new license, press the ADD NEW button and enter the license code:

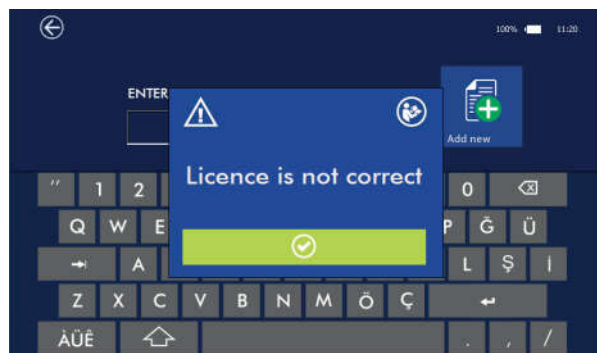




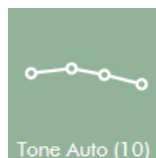
The ADD NEW button should be pressed again to confirm the device recognizes the validity of the license and will display a confirmation message stating the test that has been added



An error is displayed if the license is not valid.



If there are no active licenses, the tests can be used in demonstration mode up to 10 times.



The number in brackets shows the number of demonstrations remaining.

Date and time



Press the **DATE** icon (see adjacent image) to set date and time.



Use the arrows to adjust the date and time.

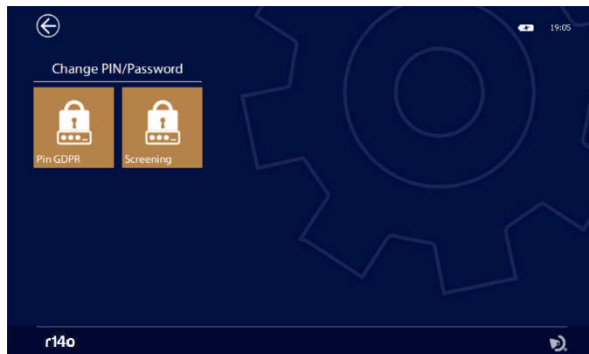


Press this icon to confirm.

Security



Opens the security page.



Allows setting the GDPR pin code which is used at startup of the instrument



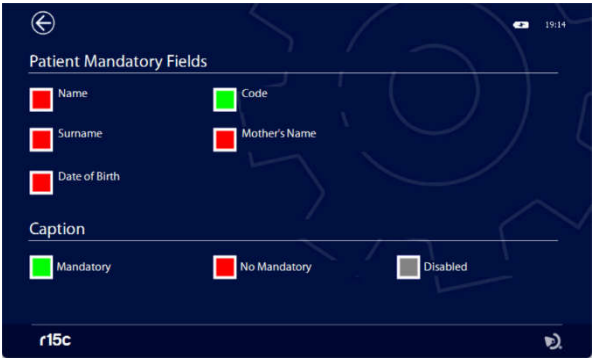
Allows setting the Screening pin code which is used to grant access to the setup in Screening TEOAE and Screening DPOAE.

Patient Mandatory Fields



Mandatory

Allows setting what fields of the patient database are obligatory. The Mother's name is specific for newborn screening programs and can be disabled / hidden.



About



About

Pressing the **ABOUT** icon displays the devices' serial number, firmware, and software versions.



5.2 OAE Setup

Press the **OAE** tab to enter the respective setup page:



End of test criterion after obtaining the PASS



With this key (see adjacent image) it is possible to establish the criterion for ending the test in case of Pass: it is possible to end the test just when the Pass is obtained or, even if the pass was obtained, continue to the end of time and see overall result.



DP-OAE test frequency scan



Pressing this button (see adjacent image) changes the scanning order of the DP test frequencies: from the minor to the major or vice versa.

OAE Probe check



This function allows you to check the operation of the OAE probe. Press the key to access the verification menu, it is necessary to insert the probe in the test cavity provided to start the verification.



The functionality will be checked and the result will be provided on the screen; if OK, the tests can be carried out, otherwise the probe must be checked for: tip cleaning; tip correctly inserter into probe body; front part probe coupling (both brackets hooked), see chap. 11.1 for more information.



If the malfunction persists, contact technical support.

Extension of the maximum ear volume limit



The device automatically adjusts the levels of stimuli adapting to the volume of the ear, the maximum limit is about 2.8ml. In some cases, it may be necessary to perform the test even if a higher volume is detected (e.g., perforated eardrum). With this key it is possible to extend the test up to volumes of about 4ml; in the latter case the levels will be generated as if you were in the presence of a 2ml volume.

Minimum OAE level in DP test



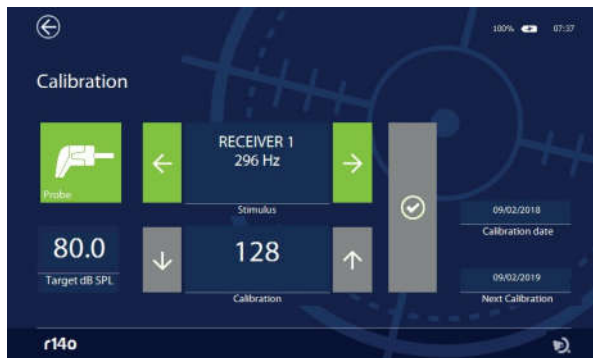
In the DP test the OAE must be higher than a minimum threshold level to be considered as valid and then proceed to evaluate the signal to noise ratio; this level can be set using this button at -5, -10 or -15dB SPL.

5.3 OAE Calibration



Entering the OAE setup menu and pressing the **CALIBRATION** icon (see adjacent image), you will be able to access to the calibration page.

Only authorized personnel are allowed to perform a new calibration.



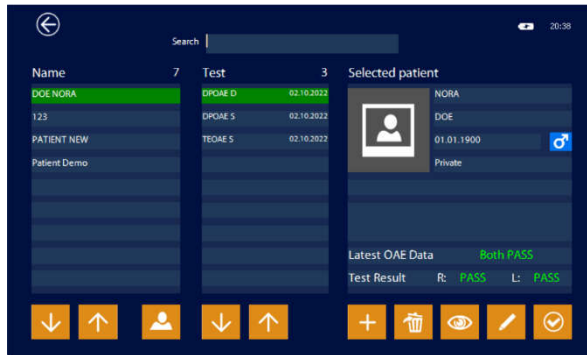
The device calibration must be performed by trained technical staff only. Information concerning how to perform a calibration are available on the Technical Manual.

6. Patient database management



The device has an internal memory which can store patient data and test results. The data-base is accessed by pressing the **PATIENT** button on the home screen.

In the database, all data stored can be displayed.



Database

The left column lists patients, the center column shows test sessions associated to the patient, and the right-hand column displays demographics.



Using the **ARROW** keys to navigate between patients, results for the highlighted patient will be displayed



Patients list can be sorted by name or company name (if inserted in the patient data). Press the icon shown on the left to change the sorting option.

Patient data

The right side of the screen displays the selected patient's personal data. The lower part contains some icons that activate the specific functions described below.



Create new patient. Use the + button to add a new patient to the database.



To display the accented characters, simply press and hold the key for a few seconds: a pop-up will appear with the alternatives (e.g., hold down the A key to enter Â, Á, Ã, Ä, ...) Using the options key, you can select different keyboard layouts with the corresponding character sets (e.g., Qwerty, Arabic, Cyrillic ...)



Patient data deletion. Pressing the **DELETE** icon will delete the highlighted test. Once all tests have been deleted, the patient can be removed from the database by pressing **DELETE** again.



Patient data editing. Pressing the **EDIT** icon allows data editing for the selected patient.



Test results display. Press the **VIEW** icon (see adjacent image) to display a specific test result for the selected patient.



Patient activation. Press the icon shown on the left to activate the selected patient. The selected patient's name will be displayed on the top left of the display once you enter one of the pages that allow the tests execution. Tests that have been performed will be saved to the associated patient.

7. Print management

The test results can be printed through the PC Suite Resonance MDS, or directly from the unit to the optional portable wireless printer.

To print out via the MDS, Transfer the data from the device to the PC Suite Resonance MDS and then follow the instructions related to the "Session report" MDS function.

If you wish to print directly from the unit, switch the printer on and insert the WiFi dongle in the R14 OUSB port. Please note that the two devices require 20-30 seconds to connect.



Once unit and printer are connected, enter the patient database menu, select the test you wish to print out and press the **PRINT** icon you will see on the display. It is possible to print report during the text exit by pressing the print button on the pop up menu.



The wireless printer uses the 2.4 GHz Wi-Fi technology. Using it in areas where too many Wi-Fi networks, or other sources of electromagnetic interference such as mobile phones or other similar devices can be found, can provoke malfunctions.

8. Data transmission to PC

Data transfer from the device to PC is possible via Bluetooth connection.

To transfer test results stored within the R140 and subsequently view, edit, print or store them in a NOAH database, the PC suite Resonance MDS must be installed on to a computer. The installation file named **MRS Resonance Setup.exe** is supplied on the USB pen-drive which includes this operating manual.

Refer to the installation & operating instructions provided with the PC suite Resonance MDS for further details.

How to associate r14o to PC via Bluetooth

1. Activate the Bluetooth connection on the device by pressing Bluetooth button present in the General Setup page (see section 5.1 of this manual).
2. Make sure the Bluetooth data transmission is active on your PC.
3. Open **Windows settings** on PC.
4. Select the **Devices** icon.
5. Open **Bluetooth** devices management feature.
6. Wait until your PC detects the device. This may take several minutes. When finished, an icon showing the serial number of the device to be associated will be displayed.
7. Check that the serial number displayed corresponds to the one on the device identification label. If it matches, press the icon on screen to complete the association.
8. The unit will generate a **PASSCODE** which you will see displayed on the screen.
9. The **PASSCODE** will be displayed on the PC screen, too.
10. If the two codes are the same, press the confirm button on the unit first and then on the PC.
11. When this process is finalized, the unit is associated.

As an alternative, data can be transferred via cable through the unit mini-USB port. If you choose this option, be aware that by connecting a Personal Computer to a Medical Device you create a system which, if used in a patients' area need to comply with all safety requirements.



In case the r14o is connected to other devices equipped with their own external power supplies, the compliance of the whole system to the IEC 60601-1 standard needs to be verified under the technician's connecting the unit responsibility. In case insulation devices are used, they have to comply with the IEC 60601-1 standard.

Each person connecting an r14o to other external units creates a system and is held responsible for the compliance of the system itself to the IEC 60601-1 standard.

9. Main labels

Device labels

TOP HOUSING



RIGHT SIDE



FRONTAL VIEW

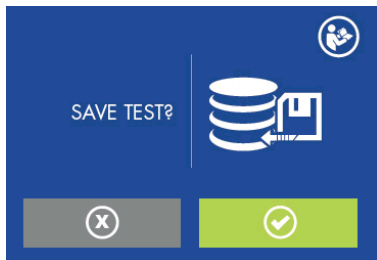


LEFT SIDE



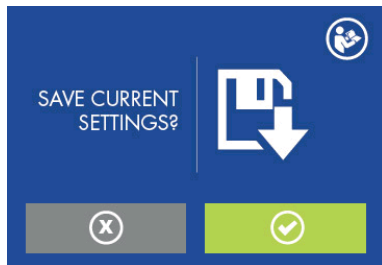
BOTTOM HOUSING

10. Messages and warnings

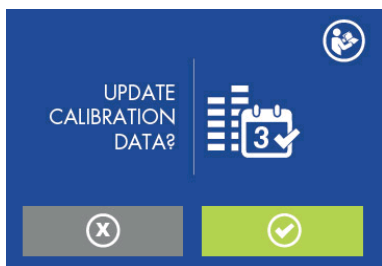


Save the test in the database?

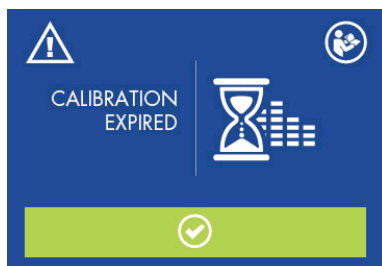
Is it possible to perform the print out. The button is active only if the printer is active and connected to the Wi-Fi adapter.



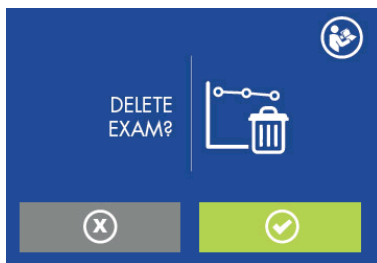
Save current settings in setup?



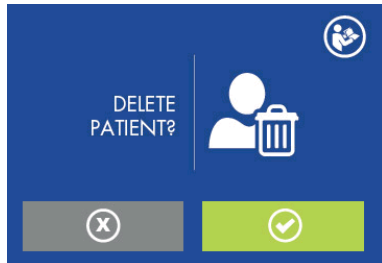
Update also the calibration date?



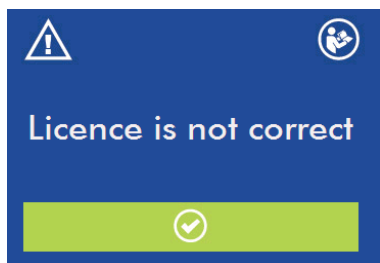
Calibration expired. Recalibration required. Continue anyway?



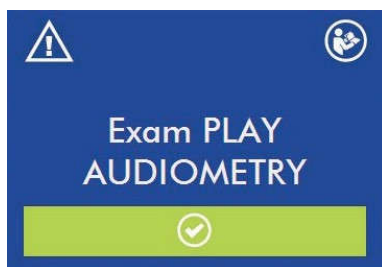
Delete the exam from database?



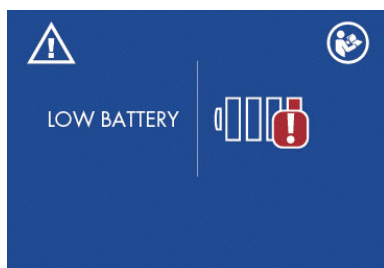
Delete the patient from database?



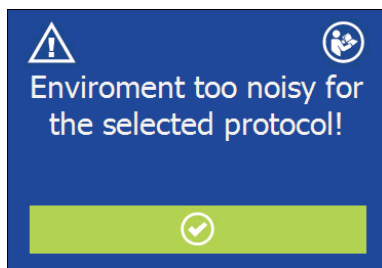
Attention, optional test license is not correct.



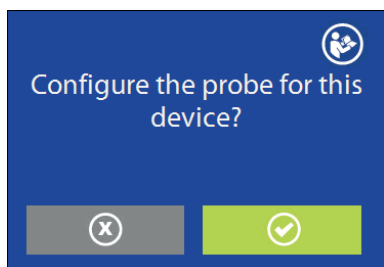
Optional test license is correct.



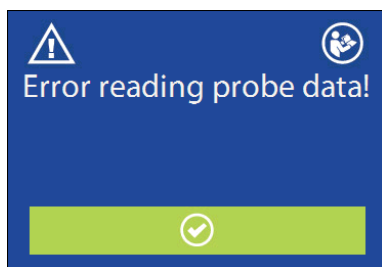
Attention, battery is running low. Connect the device to its medical power supply to recharge and continue the test.



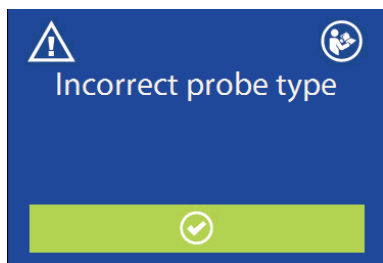
The noise level is too high for the test to be performed. Check the probe correct positioning and/or the selected noise level in the Setup menu.



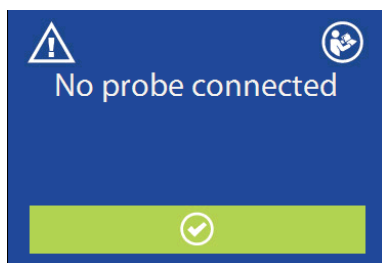
The connected probe has no serial number. You need to confirm whether you want to be permanently coupled to this specific device.



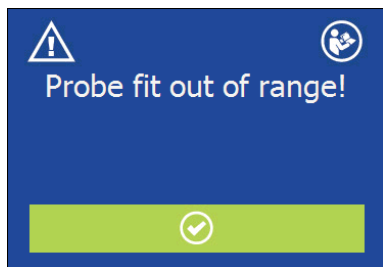
An error occurred while reading the probe data. Check the probe connector.



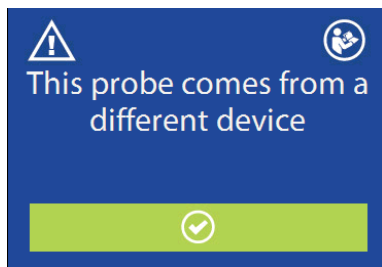
The connected probe is not suitable for the selected test. The probe needs to be replaced with the correct one.



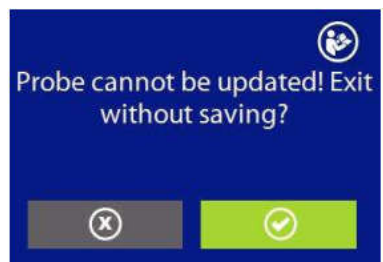
No probe is connected. Connect a probe.



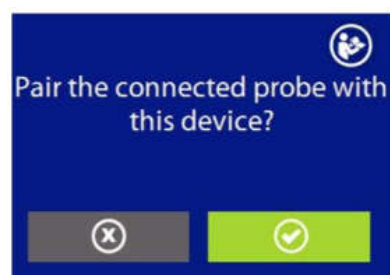
The probe is not properly inserted into the ear. Check the tip size.



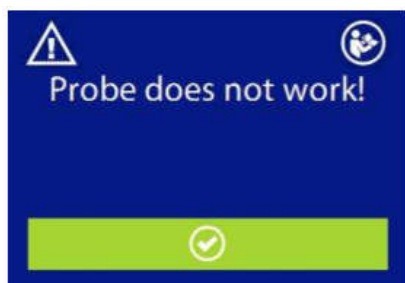
The connected probe is coupled to another device. The test can be run anyway.



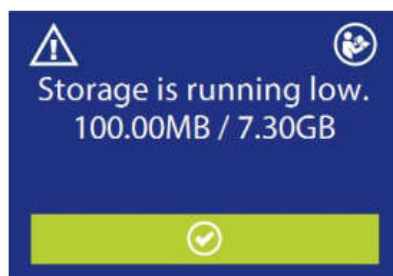
The internal probe memory cannot be updated with the calibration data! Exit without saving?



Pair the connected probe with this device? This operation will configure the probe to work permanently with the device



Probe does not work!



The free memory for saving exams is running out. It is possible to free up memory by transferring the data to the MDS suite and deleting them from the device

11. Care and maintenance



Switch off the device and disconnect the power plug before cleaning!

Always disconnect the mains conductor during the cleaning process, and be careful that no fluid enters inside of the device or the accessories; no alcohol or spirits should be used.

The device requires cleaning the housing with a damp cloth (do not use aggressive cleaning agents); avoid liquid ingress.

During testing operations, probe tips come into direct contact with the patient. Although said components are biocompatible, they are not sterile and they must therefore be subject to special care:

The probe tips are disposable: use only once and substitute them before use with a new patient.

Calibration and any possible repair, installation and/or update of the device must be strictly carried out only by specialized technical personnel, duly authorized by the r14o manufacturer and in full compliance with the terms and conditions specified in the Technical Manual.

Standard maintenance and device calibration operations are to be regularly performed at one-year intervals. Standard maintenance operations cover the inspection of the transducer cable conditions, of the mains power supply cable, as well as an inspection check to ensure the device's outer section and/or cabinet are in good working order. All device calibration operations must be strictly performed only by qualified technical personnel, duly authorized by the manufacturer and implementing the appropriate apparatus for checking the levels and frequencies of all the output signals by the transducers supplied with the device. Once device calibration testing using the special purpose apparatus is complete, it is necessary to run a device function test.

Should transducers require replacement, use only parts provided by Resonance. Upon replacement of transducers, the entire system will require re-calibration.

The device is provided with a lithium-ion battery type 1SPNCR-18650A. Said battery is fitted into the device and possible replacement thereof must strictly be performed only by qualified technical personnel. In case of replacement, use only the same battery type.

11.1 Otoacoustic emission probe maintenance

Before using the probe, make sure it is clean and its transparent plastic tip is totally free from ear wax residues to prevent blockages.

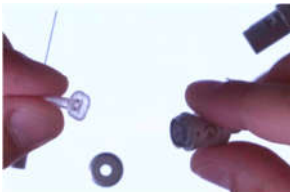
To clean the probe proceed as follows:



- 1.** Slightly but firmly hold the probe and turn the upper part to align the two black dots.



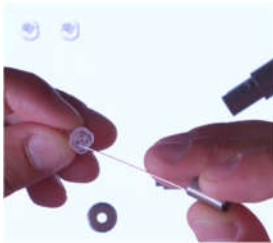
- 2.** Separate the tip of the probe from the probe.



- 3.** Pull out the transparent part of the tip.



- 4.** Clean the 3 tip channels using the supplied cleaning tool.



5. Make sure the 3 channels are completely free from any kind of debris. Eventually change the probe tip with a new one.



6. Place the transparent probe tip back making sure the recess on the tip is properly matched with the corresponding tooth on the front part of the probe. Please note that in case the tip is inaccurately positioned, closing the probe is not possible.



7. Place the front part of the probe tip back into place making sure that the two black dots are aligned. Then turn firmly the upper part to close it.

12. Safety regulations

12.1 Electrical Safety



The device applied parts comply to type BF of IEC 60601-1, in any case, the device is not intended to be used in environments dealing with explosive material or equipment.

12.2 Device handling and operation

The device should be checked once daily, as per operations described on section 11.

The device is designed to be handled and operated by trained personnel such as Audiologist, ENT doctors or personnel with similar qualifications ONLY.

12.3 Measurement Safety

As a manufacturer, we recommend to perform an annual inspection and calibration in order to ensure safety and quality of the measurement.

Annual calibration and maintenance should only be carried out by RESONANCE authorized personnel.

M.R.S. will not be liable for any failure to comply with the specified inspection date, according to the Medical product law. The use of uncalibrated and uninspected device is strictly prohibited.

13. Technical specifications

Model	r14o-TE		r14o-DP		r14o-TE/DP
Code	4300103010	4300103030	4300103015	4300103040	4300103050
	Test				
TEOAE Screener	●	●	○	○	●
TEOAE Advanced screener	○	●	○	○	●
DPOAE Screener	○	○	●	●	●
DPOAE Advanced screener	○	○	○	●	●
	Accessories				
Otoacoustic emission probe*	●	●	●	●	●
OAE kit tips*	●	●	●	●	●
Calibration cavity	●	●	●	●	●
USB-Pen Drive with User Manual	●	●	●	●	●
Medical CE approved power supply	●	●	●	●	●
Cradle	●	●	●	●	●
MDS software	●	●	●	●	●
Carrying Bag	●	●	●	●	●
External USB Keyboard	○	○		○	○
Internal Bluetooth module 2.1/4.0	●	●	●	●	●
USB cable	○	○		○	○

● = standard ○ = optional - = not available * = applied parts

Applied parts are constituted by the transducers that convey the test signals. Said transducers are applied so that they go into direct contact with the patient. In the foregoing list, they are highlighted in bold script and are counter-marked with an asterisk (*).

Reference standard:	Otoacoustic emission: IEC 60645-6 Type 2
Safety:	IEC 60601-1 (2006) + A1 (2012) Classe II Tipo BF
EMC:	IEC 60601-1-2 (2014)
Medical CE Marks:	Class IIa The Manufacturer's Quality Management System complies to the Annex II of Medical Device Directive 93/42/EEC. The notified body IMQ, Identification No. 0051, has approved the Quality System.

Otoacoustic Emissions	
Number of probes:	1
Acquisition	24 bit 44.1Ksps
TEOAE	
Stimulus level:	70 ÷ 90 dB peSPL
Stimulus frequency:	66 Hz
Stimulus type:	non-linear click
Measurement interval:	512 samples, 5 to 15 msec.
Bandwidth:	from 750 Hz to 5 KHz
Artefact noise level:	from 20 to 60 dB SPL or off
Protocols:	3 user definable
Pass stop criterion:	signal/noise ratio 3, 6, 9, or 12 dB frequency definable, for 3, 4 or 5 frequencies
Test frequency bands:	1, 1.5, 2, 3, 4 KHz
Max test time:	from 500 to 1250 stimuli (from 30 to 75 sec.)
DPOAE	
F2/F1 frequency ratio	1.22
Stimulus level L2:	35 - 70 dB SPL step size 5 dB
Stimulus level L1:	user's definable: automatic ($L1 = 0.4 * L2 + 39$ dB), $L1 = L2$, $L1 = L2 + 10$ dB
Artefact noise level:	20 - 40 dB SPL
Pass stop criterion:	signal/noise ratio 3, 6, 9, or 12 dB frequency definable, for 2 up to the number of enabled frequencies
Test time per frequency:	automatic or user definable, 2 to 16 sec
Protocols:	1 fixed (65-55 dB) 3 user definable
DPOAE Screener	
Reference frequency:	F2, 1.5 - 6 kHz, 6 frequencies Max, user definable
DPOAE Advanced screener	
Reference frequency:	F2, 1.5 - 12 kHz, 15 frequencies Max, user definable

Technical descriptions more detailed and complete as wiring diagrams, parts list, instructions for calibration and other materials necessary for the repair and maintenance of the instrument, are contained in the service manual which will be provided, upon request, technical personnel authorized by the manufacturer to carry out such operations.

Dimensions	
W x D x H: 180 x 135 x 41 mm	Weight: 572 g net
Environmental conditions	
Transport and storage:	Temperature: from -20 to +50 °C
Humidity: from 20 to 90% without condensation Pressure: from 500 hPa to 1060 hPa	
Operation:	Temperature: from +15 °C to +35 °C
Humidity: from 20 to 90% without condensation Pressure: from 700 hPa to 1060 hPa	
Power	
Batteries: battery on/off switching	Internal rechargeable Lithium-Ion battery 3,6 V – 3,1 Ah Automatic
Battery status indication	
Battery duration:	Normal use, up to 8 h;
Continuous use (test performance): 3 h	
Battery charge time:	Less than 4 h; 80% in 2 h
External power supply:	Medical power supply input 100/240 V AC 50–60 Hz – output +5 V DC / 2 A
Construction	
Self-extinguishing plastic ABS cabinet	

14. Electromagnetic compatibility

Guidance and manufacturer's declaration – Electromagnetic emissions		
The R14O is suitable for use in the specified electromagnetic environment. The purchaser or user of the R14O should assure that it is used in an electromagnetic environment as described below:		
Emissions test	Compliance	Electromagnetic Environment
RF emissions CISPR 11	Group 1	This R14O uses RF energy only for its internal function. Therefore, the RF emission is very low and not likely to cause any interference in nearby electronic equipment. This R14O is suitable for use in all establishments, including domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
RF emissions CISPR 11	Class B	
Harmonic emissions IEC 61000-3-2	Complies	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Complies	

Guidance and manufacturer's declaration – Electromagnetic immunity			
The R14O is suitable for use in the specified electromagnetic environment. The purchaser or user of the R14O should assure that it is used in an electromagnetic environment as described below:			
Immunity Test	IEC 60601-1-2 Test level	Compliance level	Electromagnetic Environment
Electrostatic discharge (ESD) IEC 61000-4-2	8 kV contact 2/4/8/15 kV air	IEC 60601-1-2 Test level	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Radiated electromagnetic field IEC 61000-4-3	3V/m 80 MHz to 2.7 GHz	IEC 60601-1-2 Test level	Portable and mobile RF communications equipment should be used no closer to any part of the R14O including cables. Minimum distance 30 cm.

Electrical fast transient/burst IEC 61000-4-4	2 kV for power supply lines 1 kV for input/output lines > 3 m	IEC 60601-1-2 Test level	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	0.5/1 kV differential mode 0.5/1/2 kV common mode	IEC 60601-1-2 Test level	Mains power quality should be that of a typical commercial or hospital environment.
Conducted disturbances induced by RF fields IEC 61000-4-6	3 V 150 kHz to 80 MHz	IEC 60601-1-2 Test level	Portable and mobile RF communications equipment should be used no closer to any part of the R14O including cables. Minimum distance 30 cm.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	0% Un for 0.5 cycle 0% Un for 1 cycle 70% Un for 25 cycles 0% Un for 5 s	IEC 60601-1-2 Test level	Mains power quality should be that of a typical commercial or hospital environment. If the user of the R14O requires continued operation during power mains interruptions, it is recommended that the R14O be powered from an uninterruptible power supply or a battery.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	30 A/m	IEC 60601-1-2 Test level	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

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